

LINFO2347: Network Attacks

Project 2: Attacks and defenses on Mininet

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1. Port scan

RED

- TCP SYN + UDP scan using **Scapy**
- SYN-ACK = open, RST = closed, no reply = filtered
- Configurable port range, targets, timeout

BLUE

- Per-IP rate limiting on r2 (meter, 15 SYN/s max)
- UDP from internet to DMZ rejected with ICMP port-unreachable
- Large scans get most ports dropped, small scans still work

2. SSH brute-force

attack

- Tries username/password combinations using **Paramiko**
- Reads from wordlists (users.txt, passwords.txt)
- Stops on first valid credentials found

defense

- Per-IP rate limit on port 22: max 2 new connections/minute
- Uses nftables meter on both r1 and r2
- After a few attempts, attacker gets timed out

3. FTP brute-force

attack

- Same approach as SSH, using **ftplib** on port 21
- Iterates over all user/password combinations
- Stops on first successful login

defense

- Per-IP rate limit on port 21: max 2 new connections/minute
- Same meter mechanism as SSH protection
- Loaded on both r1 and r2

4. ARP cache poisoning

attack

- Sends forged ARP replies from ws2 using **Scapy**
- Tells ws3: "r1's IP is at my MAC"
- Tells r1: "ws3's IP is at my MAC"
- Result: MITM, all traffic goes through ws2

defense

- nftables ARP table on ws3
- Drops ARP replies claiming r1's IP with wrong MAC
- MAC must be updated per Mininet session

5. DNS reflected DoS

attack

- Spoofed DNS queries sent to dns server (port 5353)
- Source IP = victim (ws2), destination = DNS server
- DNS replies flood ws2 with amplified responses

defense

- Base firewall on r2 blocks this by default
- Spoofed source (10.1.0.2) does not match allow rule (10.2.0.0/24)
- Packets dropped before reaching the DNS server

Thank you

Any questions?